Square-root restarts and Mazur density

Conditional closure criteria for ordinary Collatz

Collatz-Final project
github.com/randogelist/Collatz-Final

Abstract

We reconstruct three conditional closure criteria from the supplied dagger–Mazur Lean developments. Square-root restarts partition hypothetical bad starts into terminal fibres and failure-depth layers. A direct estimate for each fixed predecessor basin shows that cofinal smallness of the maximal layer mass forces zero lower harmonic density of that basin. Mazur's fixed-target positive predecessor-density theorem then yields Collatz. A second route uses existence of the harmonic density of each fixed terminal fibre; geometric gaps force every such limit to vanish. A synchronized weighted aggregate gives a third formulation. The analytic smallness and regularity conditions remain open. We distinguish source-level proofs, a supplied historical native log, and the external density theorem retained as an axiom in one Lean package.

1 The main conditional reduction

For the ordinary map

$$C(n) = \begin{cases} n/2, & 2 \mid n, \\ 3n+1, & 2 \nmid n, \end{cases} \qquad \mathsf{Collatz}: \ \forall n \geq 1, \ \exists t \geq 0, \ C^t(n) = 1,$$

write $\mathcal{B} = \{n \geq 1 : \forall t, \ C^t(n) \neq 1\}$ and $\mathcal{P}_c = \{n \geq 1 : \exists t, \ C^t(n) = c\}$. The density input is

$$\mathsf{M}: \quad \forall c \geq 1, \ 3 \nmid c, \quad \exists \delta_c > 0, \ X_c, \ \forall X \geq X_c, \quad \#(\mathcal{P}_c \cap [1, X)) \geq \delta_c X. \tag{M}$$

This is the statement of Mazur's Theorem 1.1 [1]; the constants may depend on c and no convergence assumption on c is made.

Let $H_N = \sum_{n=1}^{N} 1/n$, let $p_c(N) = H_N^{-1} \sum_{n \leq N, \ n \in \mathcal{P}_c} 1/n$, and let D_N be the maximal normalized mass of an actual failure-depth layer, defined in Section 4. For every bad c and $N, M \geq 1$ the source derives

$$\boxed{p_c(N) \leq M D_N + \frac{2\log(2c)}{\log 2}\, 2^{-M} + \frac{1}{H_N}.} \tag{1}$$

Theorem 1.1 (Direct basin closure)**.** *With* $\mathsf{F}: \liminf_{N\to\infty} D_N = 0$*, one has*

$$\boxed{\mathsf{M} \implies (\mathsf{F} \iff \mathsf{Collatz}).}$$

The forward proof fixes M before selecting a large favorable N. It requires neither a global terminal-tail estimate nor a Tao theorem. No proof of F is supplied.

2 Actual square-root restarts and finite terminal packets

For $n > 1$, set

$$\tau(n) = \min\{t \geq 0 : C^t(n)^2 \leq n\}, \qquad J(n) = \begin{cases} C^{\tau(n)}(n), & \text{if this set is nonempty,} \\ 0, & \text{otherwise.} \end{cases}$$

Set $J(0) = 0$ and $J(1) = 1$. The value 0 records a failed crossing. These are classical, noncomputable definitions: a finite orbit timeout does not establish failure.

Proposition 2.1 (First-crossing geometry)**.** *If $n > 1$ and $J(n) = r > 0$, then*

$$r^2 \leq n < 4r^2, \qquad r < n, \qquad \exists t,\ C^t(n) = r. \tag{2}$$

Proof. The first inequality is the crossing condition. Immediately before the first crossing, the orbit is above $\sqrt{n}$. Its crossing step cannot be an odd step, which increases a positive value. It is a halving step from $2r$, so $n < (2r)^2$. Strict decrease follows from $r^2 \leq n$ and $n > 1$. □

Let $E = \{e > 1 : J(e) = 0\}$ be the failed endpoints. For $e \in E$, define

$$F_{e,k} = \{n \geq 1 : J^k(n) = e\}, \qquad F_e = \bigcup_{k \geq 0} F_{e,k}.$$

Before arriving at e, these iterates are successful restarts. Badness is preserved by a successful restart. Their strict decrease implies that each bad start reaches a unique failed endpoint after a unique finite number of successful restarts. Consequently,

$$\mathcal{B} = \bigsqcup_{e \in E} \bigsqcup_{k \geq 0} F_{e,k}, \qquad e \in E \Rightarrow e \geq 5. \tag{3}$$

The small cases $1, 2, 3, 4$ reach 1. Induction in (2) gives

$$n \in F_{e,k} \quad \Longrightarrow \quad e^{2^k} \leq n, \qquad 4n \leq (4e)^{2^k}. \tag{4}$$

In particular, every fixed-depth cell is finite.

Proposition 2.2 (Fixed-basin finite cover)**.** *For a bad target c,*

$$\mathcal{P}_c \subseteq \bigcup_{\substack{e \in E \\ e < c^2}} F_e. \tag{5}$$

Proof. Let $m(n) = \min_{t \geq 0} C^t(n)$. A successful first square-root restart occurs at or before the first occurrence of this minimum, so it preserves $m(n)$. At a failed endpoint e, one has $e < m(e)^2$. If $n \in \mathcal{P}_c$, then $m(n) \leq c$. Its terminal endpoint therefore satisfies $e < m(e)^2 = m(n)^2 \leq c^2$. □

Source: [R] `jump_square_upper`, `chain_power_bounds`, `pred_finite_cover`; [Z] `successful_restart`, `orbitMin_J`, `failure_lt_orbitMin_sq`.

3 Fixed-fibre regularity is a closure condition

For a set $A \subseteq \mathbb{N}_{>0}$, use the strict-cutoff notation

$$S_A(X) = \sum_{\substack{1 \le n < X \\ n \in A}} \frac{1}{n}, \qquad H(X) = \sum_{1 \le n < X} \frac{1}{n}, \qquad d_A(X) = \frac{S_A(X)}{H(X)}.$$

The empty initial cutoffs are immaterial. Define

$$\mathsf{R}: \quad \forall e \ge 2,\ \exists \ell_e \in \mathbb{R}, \quad d_{F_e}(X) \longrightarrow \ell_e. \tag{R}$$

For $e \notin E$ take $F_e = \varnothing$. This premise asks only for existence of each fixed-fibre limit.

Lemma 3.1 (Geometric gaps force a zero limit)**.** *For every $e \in E$, any existing limit of d_{F_e} equals zero.*

Proof. Set $U_r = (4e)^{2^r}$ and $V_r = e^{2^{r+1}}$. Since $e \ge 5$, $U_r < V_r$. If $n \in F_{e,k}$ and $k \le r$, (4) implies $n < U_r$; if $k \ge r+1$, it implies $n \ge V_r$. Thus $F_e \cap [U_r, V_r) = \varnothing$ and $S_{F_e}(U_r) = S_{F_e}(V_r)$. Moreover,

$$\frac{H(U_r)}{2^r} \longrightarrow \log(4e), \qquad \frac{H(V_r)}{2^r} \longrightarrow 2\log e.$$

If $d_{F_e} \to \ell_e$, dividing the equality of masses by 2^r gives $\ell_e \log(4e) = 2\ell_e \log e$. Since $4e < e^2$, the logarithms differ and $\ell_e = 0$. □

Lemma 3.2 (Natural density supplies harmonic mass)**.** *If $\#(A \cap [1, X)) \ge \delta X$ eventually for some $\delta > 0$, then $d_A(X) \ge \eta$ eventually for some $\eta > 0$.*

Proof. Writing $A_{\le}(m) = \#(A \cap [1, m])$, finite summation by parts gives

$$\sum_{\substack{n \le N \\ n \in A}} \frac{1}{n} = \frac{A_{\le}(N)}{N} + \sum_{m=1}^{N-1} \frac{A_{\le}(m)}{m(m+1)}.$$

The eventual linear counting bound gives a lower bound $\delta \log N - O(1)$. Divide by $H_N = \log N + O(1)$ and take, for example, any fixed $\eta \in (0, \delta)$ after increasing the cutoff. □

Theorem 3.3 (Regularity closure)**.**

$$\boxed{\mathsf{M} \implies (\mathsf{R} \iff \mathsf{Collatz}).} \tag{6}$$

Proof. A failure of Collatz yields an odd bad value by repeated halving. If it is divisible by 3, its next iterate is 1 modulo 3 and remains bad. Thus there is a bad c with $3 \nmid c$. By (M) and Lemma 3.2, its basin has eventually positive normalized harmonic mass. By (5), that mass is at most $\sum_{e<c^2} d_{F_e}(X)$, a fixed finite sum tending to zero under R. This is a contradiction. Conversely, Collatz makes all F_e empty. □

Source: [R] `gap_regular_limit_zero`, `positive_harmonic_lower`, `DAGGER_MAZUR_REGULARITY_EQUIVALENCE`.

Only finite sums enter the proof. Gap geometry by itself does not establish the regularity premise.

4 Failure-depth layers and the terminal basin estimate

Collect all endpoints at a common depth:

$$L_k = \bigsqcup_{e \in E} F_{e,k}, \qquad A_k = \bigcup_{i<k} L_i, \qquad K(N) = \min\{k : N < 2^{2^k}\}.$$

For $N \geq 1$ define

$$\mu_N(S) = \frac{1}{H_N} \sum_{\substack{n \leq N \\ n \in S}} \frac{1}{n}, \quad d_k(N) = \mu_N(L_k), \quad D_N = \max(\{0\} \cup \{d_k(N) : k < K(N)\}).$$

From (4), no bad start $n \leq N$ has failure depth $k \geq K(N)$. Thus

$$\mu_N(\mathcal{B}) = \sum_{k<K(N)} d_k(N), \quad \mu_N(A_{k+1}) - \mu_N(A_k) = d_k(N), \quad 0 \leq D_N \leq 1. \tag{7}$$

For a fixed bad c, split its basin at $k = (K(N) - M)_+$, where $x_+ = \max(x, 0)$. The last at most M layers contribute at most MD_N:

$$p_c(N) \leq MD_N + \mu_N(\mathcal{P}_c \cap A_k). \tag{8}$$

This intersects the early part with the *fixed basin*; it does not replace it by the larger global early-failure mass.

Lemma 4.1 (Early-basin localization)**.**

$$n \in \mathcal{P}_c \cap A_k \quad \Longrightarrow \quad n < (2c)^{2^k}. \tag{9}$$

Proof. Choose its terminal endpoint $e < c^2$ and depth $i < k$. Equation (4) gives $4n \leq (4e)^{2^i}$. Since $4e < (2c)^2$ and $i + 1 \leq k$, this implies (9). For $k = 0$ the set A_0 is empty. □

Consequently,

$$\mu_N(\mathcal{P}_c \cap A_k) \leq \frac{1 + 2^k \log(2c)}{H_N}. \tag{10}$$

If $K(N) > 0$, its minimality gives $2^{2^{K(N)-1}} \leq N$, and hence

$$2^{K(N)} \log 2 \leq 2H_N. \tag{11}$$

For $1 \leq M \leq K(N)$, substitute $k = K(N) - M$ into (10)–(11):

$$\mu_N(\mathcal{P}_c \cap A_k) \leq \frac{2 \log(2c)}{\log 2} \, 2^{-M} + \frac{1}{H_N}. \tag{12}$$

For $M > K(N)$ the early mass is zero, so (12) still holds. Combining (8) and (12) proves the main finite estimate (1).

Source: [Z] `ACTUAL_BAD_MASS_EQ_LAYER_SUM`, `BASIN_WINDOW_SPLIT`, `BASIN_EARLY_SIZE_BOUND`, `K_POWER_HARMONIC_BOUND`, `DIRECT_BASIN_TERMINAL_BOUND`.

The finite support bounds are exact arithmetic statements. The sets still use true failed crossings and do not acquire a computable membership test merely because a cutoff is finite.

5 Zero lower basin density and the exact open frontier

The precise epsilon form of the remaining condition is

$$\mathsf{F}: \quad \forall \varepsilon > 0,\ \forall N_0,\ \exists N \geq N_0, \quad D_N < \varepsilon. \tag{F}$$

Since $0 \leq D_N \leq 1$, this is exactly $\liminf D_N = 0$.

Proposition 5.1 (Bad basins have zero lower harmonic density under F)**.** *For every $c \in \mathcal{B}$,*

$$\mathsf{F} \quad \Longrightarrow \quad \forall \varepsilon > 0,\ \forall N_0,\ \exists N \geq N_0, \quad p_c(N) < \varepsilon. \tag{13}$$

Proof. Given $\varepsilon > 0$, first choose one integer $M \geq 1$ such that $\frac{2\log(2c)}{\log 2} 2^{-M} < \varepsilon/3$. Choose N_1 such that $1/H_N < \varepsilon/3$ for $N \geq N_1$. Use F to choose a common $N \geq \max(1, N_0, N_1)$ with $D_N < \varepsilon/(3M)$. Equation (1) gives (13). □

This establishes a zero *lower* density. It does not assert the full limit $p_c(N) \to 0$.

Under (M), any eligible bad target has $p_c(N) \geq \eta_c > 0$ eventually, by Lemma 3.2. This contradicts (13). Every failure of Collatz has an eligible bad target, proving the forward direction of Theorem 1.1. Conversely, under Collatz every failure layer is empty, so $D_N = 0$ for every N.

The same estimate yields a useful obstruction without assuming F:

$$\left.\begin{array}{l} c \in \mathcal{B}, \quad \mathcal{P}_c \text{ has positive lower natural density} \end{array}\right\} \Longrightarrow \exists \delta > 0,\ N_0,\ \forall N \geq N_0, \quad D_N > \delta. \tag{14}$$

Indeed, choose M to make the geometric error below $\eta_c/4$ and then N to make the harmonic error below $\eta_c/4$. Equation (1) forces $MD_N > \eta_c/2$.

One common cutoff for every layer

An equivalent countable target is

$$\forall q, N_0 \in \mathbb{N},\ \exists N \geq N_0,\ \forall k < K(N), \qquad d_k(N) \leq 2^{-q}. \tag{15}$$

The same N must work for all these layers. Separate statements $\liminf_N d_k(N) = 0$ do not give (15): two alternating zero–one sequences each have liminf zero while their maximum is always one.

A proposed majorant $U(N, k)$ suffices if it dominates every actual $d_k(N)$ and satisfies (15) with U in place of d. Both requirements need proof. The source's `FrontierMajorantCertificate` stores a function and proof fields for these unbounded requirements; it is not raw finite data accepted by a terminating checker.

Source: [Z] `ACTUAL_FRONTIER_BAD_BASIN_LOWER_LOG_DENSITY_ZERO`, `ACTUAL_FRONTIER_MAZUR_ZERO_DENSITY_CLOSURE`, `BAD_BASIN_FORCES_LAYER_FLOOR`, `ACTUAL_FRONTIER_IFF_DYADIC_TARGET`.

The remaining arithmetic task is to prove (F), or a sound common-cutoff majorant establishing it.

6 A synchronized weighted alternative

For each $e \in E$ and $k \geq 0$, define the complete layer mass

$$h_{e,k} = \sum_{n \in F_{e,k}} \frac{1}{n}.$$

The sum is finite by (4), without a cutoff on orbit time. For a bad target c set

$$X_j = 2^{2^j}, \qquad Q_c(j) = \sum_{\substack{e \in E \\ e < c^2}} \sum_{k=0}^{j} h_{e,k}. \tag{16}$$

Every predecessor $n \leq X_j$ lies in this finite packet and has depth at most j. Therefore

$$\sum_{\substack{n \leq X_j \\ n \in \mathcal{P}_c}} \frac{1}{n} \leq Q_c(j). \tag{17}$$

Under (M), Lemma 3.2 and $H_{X_j}/2^j \to \log 2$ give a positive eventual lower bound for the left side of (17), divided by 2^j.

Theorem 6.1 (Synchronized aggregate closure)**.** *Define*

$$\mathsf{A}: \quad \forall c \in \mathcal{B},\ 3 \nmid c, \qquad \liminf_{j \to \infty} \frac{Q_c(j)}{2^j} = 0.$$

Then

$$\boxed{\mathsf{M} \implies (\mathsf{A} \iff \mathsf{Collatz}).} \tag{18}$$

The forward direction contradicts the preceding lower bound. The converse is vacuous because under Collatz there are no bad targets. The small checkpoint must serve the entire finite sum in (16).

Exact inverse-restart transport

For $r > 1$ put

$$\Lambda(r) = r \sum_{\substack{n > 1 \\ J(n) = r}} \frac{1}{n}.$$

Its support lies in $r^2 \leq n < 4r^2$. Partitioning $F_{e,k+1}$ by its first restart gives the identity

$$\boxed{h_{e,k+1} = \sum_{r \in F_{e,k}} \frac{\Lambda(r)}{r}.} \tag{19}$$

A sufficient open condition is: for every $e \in E$ with $3 \nmid e$, there exist $\kappa_e \in (0,1)$ and k_e such that

$$\sum_{r \in F_{e,k}} \frac{\Lambda(r)}{r} \leq 2(1 - \kappa_e) h_{e,k} \qquad (k \geq k_e). \tag{20}$$

It implies $h_{e,k}/2^k \to 0$ by geometric decay. For $3 \mid e$, every ordinary predecessor is of the form $2^j e$, so every layer has mass at most $2/e$; its normalized mass also vanishes. For any nonnegative sequence h_k with $h_k/2^k \to 0$, splitting off a fixed initial segment shows $2^{-j} \sum_{k \leq j} h_k \to 0$. The finite endpoint sum in (16) therefore tends to zero after normalization, which proves (A).

Source: [W] `basin_mass_le_aggregate`, `COLLATZ_IFF_DAGGER_AGGREGATE`, `layerMass_succ_eq_weighted`, `weighted_contraction_implies_growth`, `aggregate_tendsto_zero_of_growth`.

The identity (19) is derived in source; (20) remains a hypothesis. Neither sparse orbit minima nor a finite parity-prefix count establishes it.

7 Formal scope, provenance, and remaining work

The statements above concern the actual ordinary map and actual restart sets. We keep (M) explicit in all displayed closure theorems. No regularity, frontier-smallness, weighted-contraction proof, or accepted finite global witness is supplied by this reconstruction.

Source	Mathematical role	Evidence in supplied archive
[R]	Fixed-fibre geometry, harmonic gaps, regularity closure	FIX6 final validation pending; preparation notes report prior module passes.
[P]	Restart partition and moving-layer formulation	FIX5 runner completion not established by the supplied archive.
[W]	Weighted transport and aggregate closure	New modules marked pending native verification.
[Z]	Actual profile, direct basin bound, zero-density closure	Native PASS transcript for the package lineage; explicit literature axiom reported for public Mazur-bound endpoints.

The core [Z] theorem takes both inputs as arguments:

$$\texttt{ActualFrontier} \longrightarrow \texttt{ActualPredecessorDensity} \longrightarrow \mathsf{Collatz}.$$

The supplied transcript reports only `propext`, `Classical.choice`, and `Quot.sound` for that conditional closure and the direct basin estimate. Public wrappers that remove the density argument additionally depend on

$$\texttt{CollatzMazurLiterature.MAZUR_2026_PREDECESSOR_DENSITY}.$$

This declaration is an *axiom* in [Z], not an imported proof of Mazur's theorem. The external manuscript identifies the mathematical input; it does not change this local dependency boundary.

The native transcript is retained evidence, not a fresh certification of the rebuilt archive. The repository's current eight-module finite-witness CI target does not compile these companion source branches. No local Lean compiler was run to prepare this paper. Future Lean compilation and audits must use the CI runner. LaTeX typesetting is separate from Lean verification.

Reproducible source map

The companion source map records archive hashes and the source bundle preserves the four original ZIP byte streams. The following identifiers refer to those exact package versions:

[R] `COLLATZ_DAGGER_MAZUR_REGULARITY_LEAN_v0_1_6_FIX6`.
[P] `COLLATZ_DAGGER_PARTITION_MAZUR_LEAN_v0_1_5_FIX5`.
[W] `COLLATZ_DAGGER_WEIGHTED_CLOSURE_LEAN_v0_2_1_FIX1`.
[Z] `COLLATZ_MAZUR_FRONTIER_ZERO_DENSITY_ATIER_LEAN_v0_2_0_REBUILT`.

These source versions are distinct developments. Their evidence is not combined into a claim of a single freshly compiled Lean chain.

Reference

[1] Lech Mazur, *Positive Lower Density of Collatz Predecessors*, version 1.0, 6 September 2026, Theorem 1.1. Original manuscript (PDF). The theorem is the external input (M); its proof is not reproduced here.